AIM:- Forward pass with fixed weights of Long short term memory (LSTM)

```
import numpy as np
np.set_printoptions(suppress=True)
def sigmoid(x):
    return 1/(1+np.exp(-x))

def softmax(x)
    exp_x = np.exp(x-np.max(x))
    return exp_x/np.sum(exp_x)
def cross_entropy(y_true, y_pred)
    return -np.sum(y_true * np.log(y_pred +1e-12))
Wf = np.array([[-0.3922, -0.0009, -0.0711, -0.6524, 0.1995],
               [0.723, 0.4047, 0.1089, 0.5632, -1.0513]])
bf = np.array([[-0.7044], [-0.5538]])
Wi = np.array([[0.2833, -0.1896, 0.7952, 0.9753, 0.1119],
               [-0.0299, -0.5395, 0.0708, 0.3999, 0.847]])
bi = np.array([[1.594], [1.48]])
Wo = np.array([[0.1023, -0.1504, 1.0928, 0.7877, 0.1063],
               [0.1616, -0.7025, 0.187, 0.9171, 0.715]])
bo = np.array([[1.8517], [1.6679]])
Wc = np.array([[-0.2761, 0.3366, 1.9493, -1.5433, -0.5777],
               [-0.6505, -0.4363, -0.7741, -1.3224, 1.6044]])
bc = np.array([[-0.1707], [-0.2208]])
Why = np.array([[1.5545, 3.5032], [3.7474, -0.473], [-2.6135,
                -2.9926]])
by = np.array([[0.3001], [0.4803], [-0.7804]])
x1 = np.array([[1], [0], [0]])
x2 = np.array([[0], [1], [0]])
x3 = np.array([[0], [0], [1]])
inputs = [x1, x2, x3]
t1 = np.array([[0], [1], [0]])
t2 = np.array([[0], [0], [1]])
t3 = np.array([[1], [0], [0]])
targets = [t1, t2, t3]
hidden_size = 2
h = np.zeros((hidden_size, 1))
c = np.zeros((hidden_size, 1))
loss = 0
print("predictions:\n")
```

```
for t in range(3):
    z = np.vstack((h, inputs[t]))
    f = sigmoid(Wf @ z + bf)
    i = sigmoid(Wi @ z + bi)
    o = sigmoid(Wo @ z + bo)
    c_tilde = np.tanh(Wc @ z + bc)
    c = f * c + i * c_tilde
    h = o * np.tanh(c)
    y = Why @ h + by
    p = softmax(y)
    loss = loss + cross_entropy(targets[t], p)
    print(f"Time step {t+1} prediction:")
    print(np.round(p, 4))
    print()
print("Final Loss:", round(loss, 6))
```

Output

```
Time step 1 prediction:
[[0.0036] [0.9827] [0.0137]]
Time step 2 prediction:
[[0.0086] [0.0096] [0.9817]]
Time step 3 prediction:
[[0.9861] [0.0082] [0.0107]]
Final Loss: 0.049908
```